

Chapter 2: Second Quantization

Exercise #1: (2.8) "Coordinate Basis for Transformation Law"

$$a_{\tilde{\lambda}}^{\dagger} = \sum_{\lambda} \langle \lambda | \tilde{\lambda} \rangle a_{\lambda}^{\dagger} \quad a_{\tilde{\lambda}} = \sum_{\lambda} \langle \tilde{\lambda} | \lambda \rangle a_{\lambda}$$

$$\text{where } |\tilde{\lambda}\rangle = a_{\tilde{\lambda}}^{\dagger} |0\rangle \quad \text{and} \quad |\tilde{\lambda}\rangle = a_{\tilde{\lambda}}^{\dagger} |0\rangle$$

(pg 46) "Transformation formula"

$$a_{\tilde{\lambda}}^{\dagger} = \sum_{\lambda} \langle \lambda | \tilde{\lambda} \rangle a_{\lambda}^{\dagger} \quad a_{\tilde{\lambda}} = \int dx \langle \tilde{\lambda} | x \rangle a(x)$$

$$a_k = \int_0^L dx \langle k | x \rangle a(x) \quad \dots \text{where} \quad \langle k | x \rangle = \langle x | k \rangle^* \\ = \frac{e^{-ikx}}{\sqrt{L}}$$

$$= \int_0^L dx \langle k | x \rangle \langle x | k \rangle a_k$$

$$= \int_0^L dx \frac{e^{-ikx}}{\sqrt{L}} \frac{e^{-ikx}}{\sqrt{L}}$$

$$= \frac{1}{L} \int_0^L dx a_k$$

$$= a_k$$

Exercise #2: (pg 21) "Hamiltonian of a Harmonic Oscillator"

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{m\omega^2}{2} \hat{x}^2$$

(pg 48) "Momentum Operator"

$$\hat{p} = -i\hbar \partial_x$$

$$\hat{H} = \frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + \frac{m\omega^2}{2} \hat{x}^2$$

$$= \int d\bar{r} \langle r | \bar{r} \rangle \langle \bar{r} | a \rangle \left(\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + \frac{m\omega^2}{2} \hat{x}^2 \right) \langle \bar{r} | r \rangle \langle r | \bar{r} \rangle$$

$$= \int d\bar{r} \langle r | \bar{r} \rangle a_r^\dagger \left(\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + \frac{m\omega^2}{2} \hat{x}^2 \right) \langle \bar{r} | r \rangle a_r$$

$$= \int d\bar{r} a^\dagger(r) \left(\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + \frac{m\omega^2}{2} \hat{x}^2 \right) a(r).$$

(2.14) "One-body Hamiltonian"

$$\hat{H} = \int d\bar{r} a^\dagger(r) \left(\frac{\hat{p}^2}{2m} + V(r) \right) a(r)$$

Equation #3: (pg 55) "Wannier representation of the Hamiltonian"

$$\hat{H} = \sum_k \epsilon_k a_{k\sigma}^\dagger a_{k\sigma} = \frac{1}{N} \sum_i \sum_{i'} e^{ik(R_i - R_{i'})} \epsilon_k a_{i\sigma}^\dagger a_{i'\sigma} = \sum_{i,i'} a_{i\sigma}^\dagger t_{ii'} a_{i'\sigma}$$

$$\text{where } t_{ii'} = N^{-1} \sum_k e^{ik(R_i - R_{i'})} \epsilon_k$$

(2.24) "Block and Wannier States"

$$a_{k\sigma}^\dagger = \frac{1}{\sqrt{N}} \sum_i e^{ik \cdot R_i} a_{i\sigma}^\dagger \quad a_{i\sigma}^\dagger = \frac{1}{\sqrt{N}} \sum_k e^{-ik \cdot R_i} a_{k\sigma}^\dagger$$

Square Lattice Geometry:

$$\hat{H}_0 = \sum_{i,i'} a_{i\sigma}^\dagger t_{ii'} a_{i'\sigma}$$

$$= \sum_{i=-1}^1 \sum_{j=-1}^1 a_{i0}^\dagger t_{ij} a_{j0}$$

① Matrix:

$$\text{diag}(\hat{H}_0) = \text{diag} \begin{pmatrix} 0 & -1 & 0 \\ -1 & 0 & -1 \\ 0 & -1 & 0 \end{pmatrix} t e^{iR\chi}$$

... when $t_{ij} = \begin{cases} -t & i \neq j \\ 0 & \text{otherwise} \end{cases}$

② Eigenvalue:

$$\hat{H}_0 - \lambda = \begin{pmatrix} -\lambda & -1 & 0 \\ -1 & -\lambda & -1 \\ 0 & -1 & 0 \end{pmatrix}$$

$$= -\lambda^3 + 2\lambda$$

$$= 0$$

$$\lambda_1 = -\sqrt{2} \quad \lambda_2 = 0 \quad \lambda_3 = +\sqrt{2}$$

③ Eigenvectors:

$$\lambda_1 = -\sqrt{2}$$

$$H - \lambda_1 = \begin{pmatrix} \sqrt{2} & -1 & 0 \\ -1 & \sqrt{2} & -1 \\ 0 & -1 & \sqrt{2} \end{pmatrix} t e^{iR\chi}$$

$$(H-\lambda)X = \begin{pmatrix} \sqrt{2} & 0 & -1 \\ 0 & \sqrt{2} & -\sqrt{2} \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

$$= \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -1 \\ \sqrt{2} \\ 1 \end{pmatrix}$$

$$\underline{\lambda_3 = -\sqrt{2}}$$

$$\hat{H}_0 \lambda_3 = \begin{pmatrix} -\sqrt{2} & -1 & 0 \\ -1 & -\sqrt{2} & -1 \\ 0 & -1 & -\sqrt{2} \end{pmatrix} t e^{i k x}$$

$$(H-\lambda_3)X = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & \sqrt{2} \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

$$= \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ \sqrt{2} \\ 1 \end{pmatrix}$$

$$\underline{\lambda_2 = 0:}$$

$$H - \lambda_2 = \begin{pmatrix} 0 & -1 & 0 \\ -1 & 0 & -1 \\ 0 & -1 & 0 \end{pmatrix} t e^{ikx}$$

$$(H - \lambda_2)X = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

$$= \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$$

④ Eigenvectors are linearly independent:

$D = \text{eigenvalues}(\hat{H})$

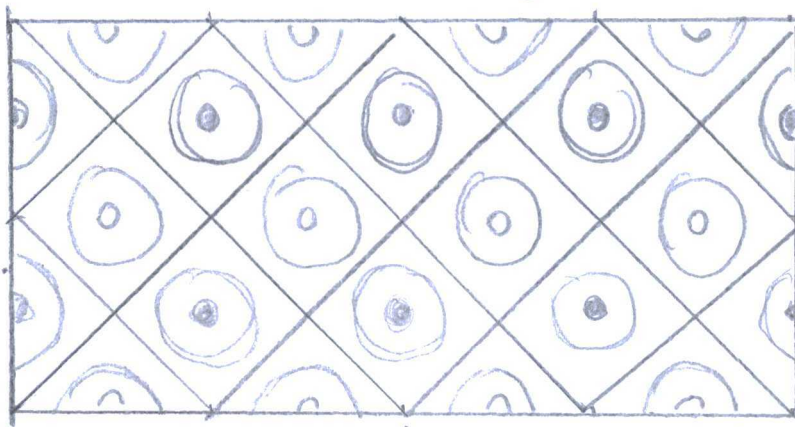
$$= \begin{pmatrix} -\sqrt{2} & 0 & 0 \\ 0 & \sqrt{2} & 0 \\ 0 & 0 & 0 \end{pmatrix} t e^{ikx}$$

Function and Contours:

$$\hat{H} = t (e^{ik_x a} + e^{-ik_x a} + e^{ik_y a} + e^{-ik_y a})$$

$$= -2t (\cos(k_x a) + \cos(k_y a))$$

$$= 0, \text{ when } k_x = k_y$$



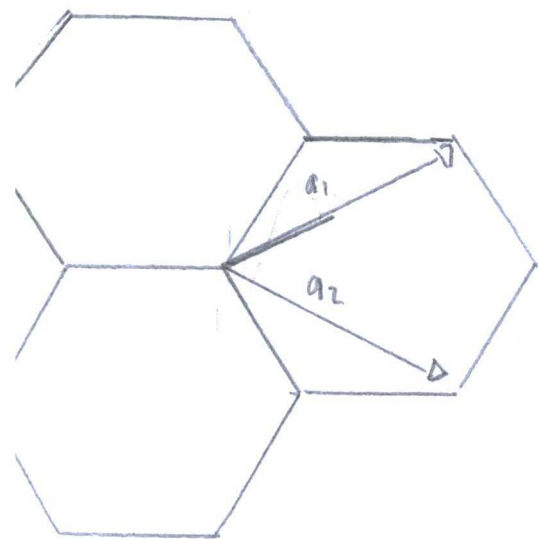
"Brillouin Zone"

Half-filling geometry separates occupancy and unoccupied bands.

Exercise #4:

Cite: Gray, D., Mookerji, B.

"Crystal structure of Graphite, Graphene, and Silicon"



$$b_1 = \frac{2\pi a_1 \times a_3}{a_1 \cdot a_3 \times a_3}$$

$$b_2 = \frac{2\pi a_3 \times a_3}{a_2 \cdot a_3 \times a_1}$$

$$= \frac{2\pi}{a} \left(\frac{1}{\sqrt{3}}, -1, 0 \right)$$

$$= \frac{2\pi}{a} \left(\frac{1}{\sqrt{3}}, 1, 0 \right)$$

$$G_{1/2} = b_1, b_2 = \frac{2\pi}{\sqrt{3}a} (1, \pm\sqrt{3})$$

"Reciprocal lattice vectors"

Fourier decomposition field operator:

(2.24) "operator transform"

$$a_{k\sigma}^\dagger = \frac{1}{\sqrt{N}} \sum_{\mathbf{R}} e^{i\mathbf{k} \cdot \mathbf{R}_i} a_{\mathbf{R}\sigma}$$

$$a_k(r) = \frac{1}{\sqrt{N}} \sum_k e^{i k R} a_{k0}$$

$$a_k(r) = \frac{1}{\sqrt{N}} \sum_k e^{-\frac{i a}{2\pi} (k_1 G_1 + k_2 G_2)} a_{k0}$$

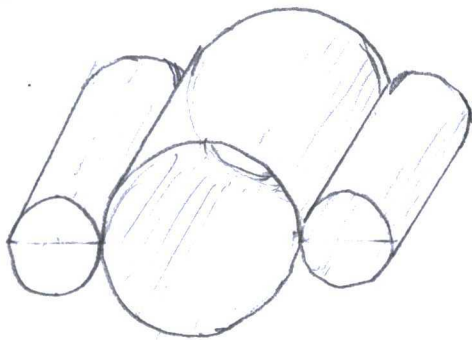
(2.25) "Fourier Representation"

$$\hat{H} = \sum_k \begin{bmatrix} a_{10}^\dagger & a_{20}^\dagger \end{bmatrix} \begin{bmatrix} 0 & -t f(k) \\ -t f^*(k) & 0 \end{bmatrix} \begin{bmatrix} a_{10} \\ a_{20} \end{bmatrix}$$

where $f(k) = 1 + e^{-i k_1 a} + e^{i(-k_1 + k_2)a}$

$$= \sum_k \begin{bmatrix} a_n^\dagger(r) & a_n^\dagger(r) \end{bmatrix} \begin{bmatrix} 0 & -t f(k) \\ -t f^*(k) & 0 \end{bmatrix} \begin{bmatrix} a_n(r) \\ a_n(r) \end{bmatrix}$$

Exercise removed from:
Edition no. 2



"armchair
graphene tube"

(2.26) "Dispersion relation"

$$E_k = \pm t |f(k)| = \pm t \sqrt{3 + 2\cos(k_1 a) + 2\cos(k_1 - k_2)a + 2\cos(k_2 a)}$$

$$= \pm t \sqrt{3 + 4\cos\left(\frac{R_1 + R_2}{2}a\right)\cos\left(\frac{R_1 - R_2}{2}a\right) + 2\cos(k_{11}a)}$$

$$= \pm t \sqrt{3 + 4\cos(0)\cos\left(\frac{R_{11}a}{2}\right) + 2\cos(k_{11}a)}$$

Trigonometric Identity - Sum to Product

$$\cos \alpha + \cos \beta = 2 \cos\left(\frac{\alpha + \beta}{2}\right) \cos\left(\frac{\alpha - \beta}{2}\right)$$

000 when $k_{\perp} = k_1 + k_2 = 0$

$$\pm \sqrt{3 + 4 \cos(k_{\parallel} a/2) + 2 \cos(k_{\parallel} a)}$$

^ (pg 58) in
edition no. 2.

Quantization rule:

(2.22) "Wannier states"

$$|\psi_{rn}\rangle = \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} e^{i\mathbf{k}\mathbf{R}} |\psi_{\mathbf{k}n}\rangle$$

$$|\psi(r + Na\sqrt{3})\rangle = \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} e^{i\mathbf{k}(r + Na\sqrt{3})} |\psi_{\mathbf{k}n}\rangle$$

$$= \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} e^{i(\mathbf{k}_1 + \mathbf{k}_2)(r + Na\sqrt{3})} |\psi_{\mathbf{k}n}\rangle$$

$e^{i\mathbf{k}}$

$$= \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} e^{i(\mathbf{k}_1 + \mathbf{k}_2)(r + L)} |\psi_{\mathbf{k}n}\rangle$$

$$e^{i(\mathbf{k}_1 + \mathbf{k}_2)(r + Na\sqrt{3})} = e^{i(\mathbf{k}_1 + \mathbf{k}_2)(r + L)}$$

$$L = Na\sqrt{3}$$

and

$$(\mathbf{k}_1 + \mathbf{k}_2)L = \frac{2\pi m}{L}$$

<u>Euler's Identity:</u> $e^{i\pi} + 1 = 0$ $e^{2\pi i} = 1$

The book has a $\sqrt{3}$ coefficient
not in the above answer. Also,
the author in second edition
supported zero-energy excitation.
Unsure about zero-energy.

Cite: R. Saito, Dresselhaus (Wife & Husband)
"Physical properties of
Carbon Nanotubes"
University of Electro-
Communications, Tokyo.
Massachusetts Institute
of Technology.

Zig-zag tubes relate to gas
and liquid sensors through
resistance (ohm) during exposure.

Exercise #6: (from other sources) "Hamiltonian with
Pauli matrices"

$$\hat{H} = \hbar v_F (\sigma_x k_x + \sigma_y k_y)$$

where $V_k =$ fermi velocity

$k =$ wave vector

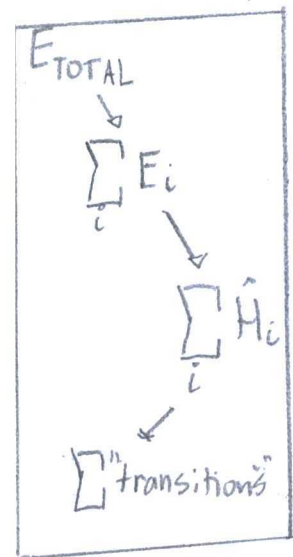
$\sigma_{xy} =$ Pauli matrices

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

Exercise #5:

(2.25) "Hamiltonian"

$$\begin{aligned} \hat{H} &= \sum_i \begin{bmatrix} a_{1i}^\dagger & a_{2i}^\dagger \end{bmatrix} \begin{bmatrix} 0 & -t f(q) \\ -t f(q)^* & 0 \end{bmatrix} \begin{bmatrix} a_{1i} \\ a_{2i} \end{bmatrix} \\ &= \underbrace{-a_{2i}^\dagger t f(q)^* a_{2i}}_{\hat{H}_1} - \underbrace{a_{1i}^\dagger t f(k) a_{1i}}_{\hat{H}_2} \end{aligned}$$



Exercise #6:

(from problem) "Pauli matrix identity"

$$\sigma_{\alpha\beta} \cdot \sigma_{\gamma\delta} = 2\delta_{\alpha\beta} \delta_{\gamma\delta} - \delta_{\alpha\delta} \delta_{\beta\gamma}$$

(from problem) "Spin operator for $1/2$ "

$$\hat{S}_i = \frac{1}{2} \sum_{\alpha, \beta} \sigma_{\alpha\beta} a_{i\alpha}^\dagger a_{i\beta}$$

$$\begin{aligned} \hat{S}_i \cdot \hat{S}_j &= (a_{i\alpha}^\dagger \sigma_{\alpha\beta} a_{i\beta} / 2) (a_{j\gamma}^\dagger \sigma_{\gamma\delta} a_{j\delta} / 2) \\ &= \frac{1}{4} a_{i\alpha}^\dagger a_{j\gamma}^\dagger (\sigma_{\alpha\beta} \cdot \sigma_{\gamma\delta}) a_{i\beta} a_{j\delta} \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{4} a_{i\alpha}^\dagger a_{j\alpha}^\dagger (2\delta_{\alpha\delta}\delta_{\beta\gamma} - \delta_{\alpha\beta}) a_{i\beta} a_{j\delta} \\
&= a_{i\alpha}^\dagger a_{j\alpha}^\dagger a_{i\beta} a_{j\delta} / 2 - a_{i\alpha}^\dagger a_{i\beta}^\dagger a_{i\beta} a_{j\delta} / 4 \\
&= -a_{i\alpha}^\dagger a_{j\beta}^\dagger a_{i\beta} a_{j\alpha} / 2 - \hat{n}_i \cdot \hat{n}_j / 4
\end{aligned}$$

(pg 60). "Pauli matrix identity"

$$\sum_{i \neq j} U_{ijji} a_{i\alpha}^\dagger a_{j\alpha}^\dagger a_{i\sigma} a_{j\sigma} = -2 \sum_{i,j} J_{ij}^F (S_i \cdot S_j + \frac{1}{4} \hat{n}_i \cdot \hat{n}_j)$$

"Energy" $\propto$ "Ferromagnetic" $\times$ ("spin" + "stat", coupling)

Exercise #7: (2.30) "Canonical Transformation to Hamiltonian"

$$\hat{H} \rightarrow \hat{H}' = e^{-t\hat{O}} \hat{H} e^{t\hat{O}} = e^{-t[\hat{O}, \cdot]} \hat{H} = \hat{H} - t[\hat{O}, \hat{H}] + \frac{t^2}{2!} [\hat{O}, [\hat{O}, \hat{H}]]$$

Second equality:

$$e^{-t[\hat{O}, \cdot]} \hat{H} = (1 - t[\hat{O}, \cdot] + \frac{t^2}{2!} [\hat{O}, [\hat{O}, \cdot]]) \hat{H}$$

$$= \hat{H} - t[\hat{O}, \hat{H}] + \frac{t^2}{2!} [\hat{O}, [\hat{O}, \hat{H}]]$$

Taylor expansion for an exponential

$$e^x = 1 + x + \frac{x^2}{2!}$$

Exercise #8: (2, 3d) "Hamiltonian expanded"

$$\hat{H}' = \hat{H}_v + \frac{t}{2} [\hat{H}_t, \hat{O}] + O(t^3)$$

(pg 62) "Ansatz"

$$t\hat{O} = [\hat{P}_s \hat{H}_t \hat{P}_d - \hat{P}_d \hat{H}_t \hat{P}_s]$$

(pg 62) "Projection Operators"

$$\begin{array}{cc} \hat{P}_s \swarrow & \hat{P}_d \swarrow \\ \text{Singly} & \text{doubly} \\ \text{occupied} & \text{occupied} \end{array}$$

$$\hat{P}_s \hat{H}' \hat{P}_s = \hat{P}_s \left(\hat{H}_v + \frac{t}{2} \hat{H}_t \hat{O} - \left(\frac{t}{2} \hat{O} \hat{H}_t \right) \right) \hat{P}_s$$

$$= \hat{P}_s \hat{H}_v \hat{P}_s + \hat{P}_s \frac{t}{2} \hat{H}_t \hat{O} \hat{P}_s - \frac{\hat{P}_s}{2} \left(\frac{\hat{P}_s \hat{H}_t \hat{P}_d - \hat{P}_d \hat{H}_t \hat{P}_s}{U} \right) \hat{H}_t \hat{P}_s$$

$$= \underbrace{\hat{P}_s \hat{H}_v \hat{P}_s}_{\hat{H}_v \hat{P}_s = 0} - \underbrace{\frac{\hat{P}_s \hat{H}_t \hat{O} \hat{P}_s}{2}}_{\langle \hat{P}_s | \hat{H}_t \hat{O} | \hat{P}_s \rangle = 0} - \frac{\hat{P}_s \hat{P}_s \hat{H}_t \hat{P}_d \hat{H}_t \hat{P}_s}{2U} + \frac{\hat{P}_s \hat{P}_d \hat{H}_t \hat{P}_s \hat{H}_t \hat{P}_s}{2U}$$

$$\hat{H}_v \hat{P}_s = 0 \quad \langle \hat{P}_s | \hat{H}_t \hat{O} | \hat{P}_s \rangle = 0$$

" $\hat{P}_s$ are eigenvalues to $\hat{H}_v$ " "Annihilation"

$$\hat{P}_s \hat{P}_d = 0$$

"Orthogonal vectors"

$$= - \frac{\hat{P}_s \hat{H}_t \hat{P}_d \hat{H}_t \hat{P}_s}{U}$$

$$= 0$$

Definition:

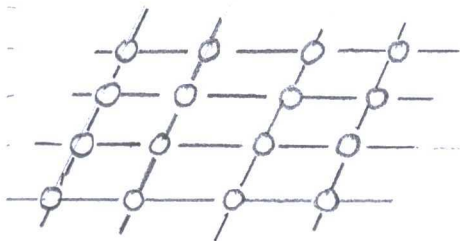
Ansatz - an assumption about a function facilitating a solution to an equation.

$$\hat{H}' = - \frac{\hat{H}_t \hat{P}_d \hat{H}_t}{U}$$

Exercise #9:

(Equation 2.29) "Hubbard Model"

$$\hat{H} = -t \sum_{\langle ij \rangle} a_{i\sigma}^\dagger a_{j\sigma} + U \sum_i \hat{n}_{i\uparrow} \hat{n}_{i\downarrow}$$



"Hubbard Model"

Where $U = \begin{cases} \infty & n=2 \\ 1 & n=1 \end{cases}$

where t = representation about state $[\pm]$ or no state $[0]$

U = Hubbard interaction for Coulomb force

(Pg 62) "Basis states"

$$\begin{aligned} |s_1\rangle &= a_{1\uparrow}^\dagger a_{2\downarrow}^\dagger |\Omega\rangle \\ |s_2\rangle &= a_{2\uparrow}^\dagger a_{1\downarrow}^\dagger |\Omega\rangle \end{aligned} \left. \vphantom{\begin{aligned} |s_1\rangle \\ |s_2\rangle \end{aligned}} \right\} \text{"singly occupied subspace"} \begin{array}{|c|c|} \hline 1 & \downarrow \\ \hline \downarrow & 1 \\ \hline \end{array}$$

$$\begin{aligned} |d_1\rangle &= a_{1\uparrow}^\dagger a_{1\downarrow}^\dagger |\Omega\rangle \\ |d_2\rangle &= a_{2\uparrow}^\dagger a_{2\downarrow}^\dagger |\Omega\rangle \end{aligned} \left. \vphantom{\begin{aligned} |d_1\rangle \\ |d_2\rangle \end{aligned}} \right\} \text{"doubly occupied subspace"} \begin{array}{|c|c|} \hline 1\downarrow & \\ \hline & 1\downarrow \\ \hline \end{array}$$

The Hubbard model describes lowest occupied atomic orbitals in ground state, with an index to each state. Hunds rule

Aufbau and Pauli exclusion principle apply.

$$\hat{H} = -t \begin{bmatrix} a_{1\uparrow}^\dagger a_{1\uparrow} & a_{1\uparrow}^\dagger a_{2\uparrow} & a_{1\uparrow}^\dagger a_{3\uparrow} & a_{1\uparrow}^\dagger a_{4\uparrow} \\ a_{2\uparrow}^\dagger a_{1\uparrow} & a_{2\uparrow}^\dagger a_{2\uparrow} & a_{2\uparrow}^\dagger a_{3\uparrow} & a_{2\uparrow}^\dagger a_{4\uparrow} \\ a_{3\uparrow}^\dagger a_{1\uparrow} & a_{3\uparrow}^\dagger a_{2\uparrow} & a_{3\uparrow}^\dagger a_{3\uparrow} & a_{3\uparrow}^\dagger a_{4\uparrow} \\ a_{4\uparrow}^\dagger a_{1\uparrow} & a_{4\uparrow}^\dagger a_{2\uparrow} & a_{4\uparrow}^\dagger a_{3\uparrow} & a_{4\uparrow}^\dagger a_{4\uparrow} \end{bmatrix} + U \begin{bmatrix} \hat{n}_{1\uparrow} \hat{n}_{1\downarrow} \\ \hat{n}_{2\uparrow} \hat{n}_{2\downarrow} \\ \hat{n}_{3\uparrow} \hat{n}_{3\downarrow} \\ \hat{n}_{4\uparrow} \hat{n}_{4\downarrow} \end{bmatrix}$$

$$= -t \begin{bmatrix} \langle d_1 | d_1 \rangle \langle s_1 | s_2 \rangle \langle s_1 | s_3 \rangle \langle s_1 | s_4 \rangle \\ \langle s_2 | s_1 \rangle \langle d_2 | d_2 \rangle \langle s_2 | s_3 \rangle \langle s_2 | s_4 \rangle \\ \langle s_3 | s_1 \rangle \langle s_3 | s_2 \rangle \langle d_3 | d_3 \rangle \langle s_3 | s_4 \rangle \\ \langle s_4 | s_1 \rangle \langle s_4 | s_2 \rangle \langle s_4 | s_3 \rangle \langle d_4 | d_4 \rangle \end{bmatrix} + U \begin{bmatrix} \bar{n}_{1\uparrow} n_{1\downarrow} \\ n_{2\uparrow} n_{2\downarrow} \\ n_{3\uparrow} n_{3\downarrow} \\ n_{4\uparrow} n_{4\downarrow} \end{bmatrix}$$

$$= -t \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} + U \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

① Matrix

$$= \begin{bmatrix} U & -t & 0 & 0 \\ -t & U & -t & 0 \\ 0 & -t & U & -t \\ 0 & 0 & -t & U \end{bmatrix}$$

Note: 1's and 0's depend on the transition rules for fermions.

② Eigen values:

$$\hat{H} - \lambda = \begin{bmatrix} U - \lambda_1 & -t & 0 & 0 \\ -t & U - \lambda_2 & -t & 0 \\ 0 & -t & U - \lambda_3 & -t \\ 0 & 0 & -t & U - \lambda_4 \end{bmatrix}$$

$$= 0$$

$$-U t^2 (U - \lambda) + \lambda t^2 (U - \lambda) - (t^2 - (U - \lambda)^2) \cdot (U^2 - 2U\lambda + \lambda^2 - t^2)$$

$$= 0$$

$$\lambda_1 = U + \frac{t}{2}(1 - \sqrt{5}), \quad \lambda_2 = U + \frac{t}{2}(1 + \sqrt{5})$$

$$\lambda_3 = U - \frac{t}{2}(1 + \sqrt{5}), \quad \lambda_4 = U - \frac{t}{2}(1 - \sqrt{5})$$

③ Eigen vectors:

$$\lambda_1 = U - \frac{t}{2}(1 - \sqrt{5})$$

$$\hat{H} - \lambda_1 = \begin{bmatrix} -\frac{t}{2}(1 - \sqrt{5}) & -t & 0 & 0 & 0 \\ -t & -\frac{t}{2}(1 - \sqrt{5}) & -t & 0 & 0 \\ 0 & -t & -\frac{t}{2}(1 - \sqrt{5}) & -t & 0 \\ 0 & 0 & -t & -\frac{t}{2}(1 - \sqrt{5}) & -t \\ 0 & 0 & -t & -\frac{t}{2}(1 - \sqrt{5}) & -t \end{bmatrix}$$

$$(H - \lambda)X = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -\frac{1+\sqrt{3}}{2} \\ 0 & 0 & 1 & \frac{1-\sqrt{5}}{2} \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -t \\ \frac{t(1-\sqrt{5})}{2} \\ -\frac{t(1-\sqrt{5})}{2} \\ t \end{bmatrix}$$

$$\lambda_2 = 1 + \frac{t}{2}(1 + \sqrt{5});$$

$$\hat{H} - \lambda_2 = \begin{bmatrix} -\frac{t}{2}(1+\sqrt{5}) & -t & 0 & 0 \\ -t & -\frac{t}{2}(1+\sqrt{5}) & -t & 0 \\ 0 & -t & -\frac{t}{2}(1+\sqrt{5}) & -t \\ 0 & 0 & -t & -\frac{t}{2}(1+\sqrt{5}) \end{bmatrix}$$

$$(H - \lambda_2)X = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -\frac{1+\sqrt{5}}{2} \\ 0 & 0 & 1 & \frac{1+\sqrt{5}}{2} \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ X_3 \\ X_4 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} X_1 \\ X_2 \\ X_3 \\ X_4 \end{bmatrix} = \begin{bmatrix} -t \\ \frac{t(1+\sqrt{5})}{2} \\ \frac{t(1+\sqrt{5})}{2} \\ t \end{bmatrix}$$

$$\lambda_3 = U - \frac{t}{2}(1+\sqrt{5})$$

$$\hat{H} - \lambda_3 = \begin{bmatrix} \frac{t}{2}(1+\sqrt{5}) & -t & 0 & 0 \\ -t & -\frac{t}{2}(1+\sqrt{5}) & -t & 0 \\ 0 & -t & \frac{t}{2}(1+\sqrt{5}) & -t \\ 0 & 0 & -t & \frac{t}{2}(1+\sqrt{5}) \end{bmatrix}$$

$$(H - \lambda_3)X = \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & \frac{2}{1-\sqrt{5}} \\ 0 & 0 & 1 & \frac{2}{1-\sqrt{5}} \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ X_3 \\ X_4 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} t \\ \frac{t(1+\sqrt{5})}{2} \\ \frac{t(1+\sqrt{5})}{2} \\ t \end{bmatrix}$$

$$\lambda_4 = 1 + \frac{t}{2}(1-\sqrt{5})$$

$$H - \lambda_4 = \begin{bmatrix} -\frac{t}{2}(1-\sqrt{5}) & -t & 0 & 0 \\ -t & -\frac{t}{2}(1-\sqrt{5}) & -t & 0 \\ 0 & -t & -\frac{t}{2}(1-\sqrt{5}) & -t \\ 0 & 0 & -t & -\frac{t}{2}(1-\sqrt{5}) \end{bmatrix}$$

$$(H - \lambda_4)X = \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & -\frac{1+\sqrt{5}}{2} \\ 0 & 0 & 1 & -\frac{1+\sqrt{5}}{2} \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} X_1 \\ X_2 \\ X_3 \\ X_4 \end{bmatrix} = \begin{bmatrix} t \\ t \frac{(1-\sqrt{5})}{2} \\ t \frac{(1-\sqrt{5})}{2} \\ t \end{bmatrix}$$

④ Eigenvalues are linearly independent:

$D = \text{eigenvalues } (\hat{H})$

$$= \begin{bmatrix} U + \frac{t}{2}(1-\sqrt{5}) & 0 & 0 & 0 \\ 0 & U + \frac{t}{2}(1+\sqrt{5}) & 0 & 0 \\ 0 & 0 & U - \frac{t}{2}(1+\sqrt{5}) & 0 \\ 0 & 0 & 0 & U - \frac{t}{2}(1-\sqrt{5}) \end{bmatrix}$$

= "eigen states"

Strong coupling: $(U/t \gg 1)$:

$$\hat{H}_{gs} = \begin{bmatrix} U/t & -1 & 0 & 0 \\ -1 & U/t & -1 & 0 \\ 0 & -1 & U/t & -1 \\ 0 & 0 & -1 & U/t \end{bmatrix}$$

Exercise in a) (2.28) "Hubbard Model"

edition no. 2

$$\hat{H} = -t \sum_{\langle i,j \rangle \sigma} a_{i\sigma}^\dagger a_{j\sigma} + \sum_i \hat{n}_{i\uparrow} \hat{n}_{i\downarrow}$$

$$= -t \left[2 \cdot a_{2\uparrow} a_{1\uparrow} \quad 2 \cdot a_{1\downarrow} a_{2\downarrow} \quad 2 \cdot a_{2\uparrow} a_{3\uparrow} \right. \\ \left. + 2 a_{3\downarrow} a_{2\downarrow} + 2 \cdot a_{3\uparrow} a_{4\uparrow} + 2 \cdot a_{4\downarrow} a_{3\downarrow} \right] \\ + U[0]$$

ooo from "Combinations":

$$\left(\begin{matrix} \text{states} \\ \text{electrons} \end{matrix} \right) = \binom{4}{3} = \frac{4!}{3!(4-3)!}$$

ooo or pictures:

1	1	1	—
—	1	1	1
1	—	1	1
1	1	—	1
↓	↓	↓	—
↓	↓	—	↓
↓	—	↓	↓
—	↓	↓	↓

$$= \begin{bmatrix} 0 & -2t & 0 & 0 \\ -2t & 0 & -2t & 0 \\ 0 & -2t & 0 & -2t \\ 0 & 0 & -2t & 0 \end{bmatrix}$$

Eigen spectrum:

Magical Relationship

Erwin Schrodinger

"Undulatory
Theory"
(1926)

$$\hat{H} \psi = E \psi$$



Cauchy

"Indirect
root"
(1826)

$$A x = \lambda x$$

Steps: ① Matrix:

$$\hat{H} = \begin{bmatrix} 0 & -2t & 0 & 0 \\ -2t & 0 & -2t & 0 \\ 0 & -2t & 0 & -2t \\ 0 & 0 & -2t & 0 \end{bmatrix}$$

② Eigenvalues:

$$\hat{H} - E = \begin{bmatrix} -E & -2t & 0 & 0 \\ -2t & -E & -2t & 0 \\ 0 & -2t & -E & -2t \\ 0 & 0 & -2t & -E \end{bmatrix}$$

$$= 0$$

$$E_1 = t(1 - \sqrt{5}) \quad E_2 = t(1 + \sqrt{5})$$

$$E_3 = -t(1 + \sqrt{5}) \quad E_4 = -t(1 - \sqrt{5})$$

④ Eigen vectors:

$$(H-E_1)\psi = \begin{bmatrix} -t(1-\sqrt{5}) - 2t & 0 & 0 & 0 \\ -2t & -t(1-\sqrt{5}) - 2t & 0 & 0 \\ 0 & -2t & -t(1-\sqrt{5}) - 2t & 0 \\ 0 & 0 & -2t & -t(1-\sqrt{5}) \end{bmatrix} \begin{bmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \\ \psi_4 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \\ \psi_4 \end{bmatrix} = \begin{bmatrix} 1 \\ t \frac{(1+\sqrt{5})}{2} \\ t \frac{(1+\sqrt{5})}{2} \\ 1 \end{bmatrix}$$

$$(H-E_2)\psi = \begin{bmatrix} -t(1+\sqrt{5}) - 2t & 0 & 0 & 0 \\ -2t & -t(1+\sqrt{5}) - 2t & 0 & 0 \\ 0 & -2t & -t(1+\sqrt{5}) - 2t & 0 \\ 0 & 0 & -2t & -t(1+\sqrt{5}) \end{bmatrix} \begin{bmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \\ \psi_4 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \\ \psi_4 \end{bmatrix} = \begin{bmatrix} -1 \\ t \frac{(1+\sqrt{5})}{2} \\ -t \frac{(1+\sqrt{5})}{2} \\ 1 \end{bmatrix}$$

$$(H-E_3)\Psi = \begin{bmatrix} t(1+\sqrt{3}) & -2t & 0 & 0 \\ -2t & t(1+\sqrt{3}) & -2t & 0 \\ 0 & -2t & t(1+\sqrt{3}) & -2t \\ 0 & 0 & -2t & t(1+\sqrt{3}) \end{bmatrix} \begin{bmatrix} \Psi_1 \\ \Psi_2 \\ \Psi_3 \\ \Psi_4 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} \Psi_1 \\ \Psi_2 \\ \Psi_3 \\ \Psi_4 \end{bmatrix} = \begin{bmatrix} -1 \\ -t \frac{(1-\sqrt{3})}{2} \\ -t \frac{(1-\sqrt{3})}{2} \\ 1 \end{bmatrix}$$

$$(H-E_4)\Psi = \begin{bmatrix} t(1-\sqrt{3}) & -2t & 0 & 0 \\ -2t & t(1-\sqrt{3}) & -2t & 0 \\ 0 & -2t & t(1-\sqrt{3}) & -2t \\ 0 & 0 & -2t & t(1-\sqrt{3}) \end{bmatrix} \begin{bmatrix} \Psi_1 \\ \Psi_2 \\ \Psi_3 \\ \Psi_4 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \\ \psi_4 \end{bmatrix} = \begin{bmatrix} 1 \\ t \frac{(1-\sqrt{5})}{2} \\ t \frac{(1-\sqrt{5})}{2} \\ 1 \end{bmatrix}$$

⑤ Eigenvectors are linearly independent, so

$$E_1 = t(1-\sqrt{5})$$

$$E_2 = t(1+\sqrt{5})$$

$$E_3 = -t(1+\sqrt{5})$$

$$E_4 = -t(1-\sqrt{5})$$

⌘
"minimum
theoretical
spectrum"

b) $S_{\text{total}} = 1/2$

Spin configurations:

oo by combinations:

$$\left(\begin{matrix} \text{states} \\ \text{electrons} \end{matrix} \right) = \binom{4}{2} = \frac{4!}{2!(4-2)!}$$

ooo or pictures

<u>1</u>	<u>1</u>	<u>L</u>	<u> </u>	<u>1</u>	<u>1</u>	<u> </u>	<u>L</u>	<u>1</u>	<u> </u>	<u>1</u>	<u>L</u>
<u> </u>	<u>1</u>	<u>1</u>	<u>L</u>	<u> </u>	<u>1</u>	<u>L</u>	<u>1</u>	<u>L</u>	<u> </u>	<u>1</u>	<u>1</u>
<u>1</u>	<u>L</u>	<u>1</u>	<u> </u>	<u>L</u>	<u>1</u>	<u>1</u>	<u> </u>	<u>L</u>	<u> </u>	<u>1</u>	<u>1</u>
<u>1</u>	<u> </u>	<u>L</u>	<u>1</u>	<u>L</u>	<u>1</u>	<u> </u>	<u>1</u>	<u>1</u>	<u>L</u>	<u>1</u>	<u>L</u>

Hubbard Model:

$$H = -t \sum_{\langle i,j \rangle}^{4,4} a_{i\sigma}^\dagger a_{j\sigma} + U \sum_i \hat{n}_{i\uparrow} \hat{n}_{i\downarrow}$$

$$= -t \sum_{\langle i,j \rangle}^{4,4} (a_{i\uparrow}^\dagger a_{j\uparrow} - a_{i\downarrow}^\dagger a_{j\downarrow} - a_{i\uparrow}^\dagger a_{j\downarrow} - a_{i\downarrow}^\dagger a_{j\uparrow})$$

$$+ U \sum_i (\hat{n}_{i\uparrow} \hat{n}_{i\downarrow} + \hat{n}_{i\downarrow} \hat{n}_{i\uparrow})$$

$$= -t (2(a_{1\uparrow}^\dagger a_{2\uparrow} + a_{1\downarrow}^\dagger a_{2\downarrow} + a_{1\uparrow}^\dagger a_{2\downarrow})$$

$$+ 2(a_{2\uparrow}^\dagger a_{1\uparrow} + a_{2\downarrow}^\dagger a_{1\downarrow} + a_{2\uparrow}^\dagger a_{1\downarrow})$$

$$+ 2(a_{2\uparrow}^\dagger a_{3\uparrow} + a_{2\downarrow}^\dagger a_{3\downarrow} + a_{2\uparrow}^\dagger a_{3\downarrow})$$

$$+ 2(a_{3\uparrow}^\dagger a_{2\uparrow} + a_{3\downarrow}^\dagger a_{2\downarrow} + a_{3\uparrow}^\dagger a_{2\downarrow})$$

$$+ 2(a_{3\uparrow}^\dagger a_{4\uparrow} + a_{3\downarrow}^\dagger a_{4\downarrow} + a_{3\uparrow}^\dagger a_{4\downarrow})$$

$$+ 2(a_{4\uparrow}^\dagger a_{3\uparrow} + a_{4\downarrow}^\dagger a_{3\downarrow} + a_{4\uparrow}^\dagger a_{3\downarrow}))$$

$$+ U[0]$$

$$= \begin{bmatrix} 0 & -6t & 0 & 0 \\ -6t & 0 & -6t & 0 \\ 0 & -6t & 0 & -6t \\ 0 & 0 & -6t & 0 \end{bmatrix}$$

Eigenspectrum:

① Matrix:

$$\hat{H} = \begin{bmatrix} 0 & -6t & 0 & 0 \\ -6t & 0 & -6t & 0 \\ 0 & -6t & 0 & -6t \\ 0 & 0 & -6t & 0 \end{bmatrix}$$

② Eigenvalues:

$$\hat{H} - E = \begin{bmatrix} -E & -6t & 0 & 0 \\ -6t & -E & -6t & 0 \\ 0 & -6t & -E & -6t \\ 0 & 0 & -6t & -E \end{bmatrix}$$

$$E_1 = 3t(1 - \sqrt{5}) \quad E_2 = 3t(1 + \sqrt{5})$$

$$E_3 = -3t(1 + \sqrt{5}) \quad E_4 = -3t(1 - \sqrt{5})$$

③ Eigenvectors:

"... a time limit"

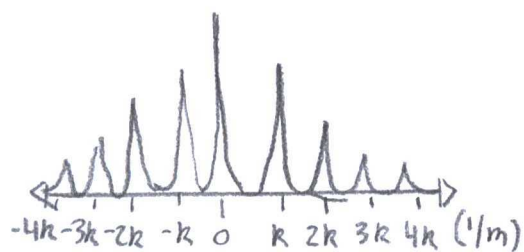
④ Eigenvectors are linearly independent:

Eigenspectrum are the eigenvectors above.

Exercise #10:

(2.35) "Hamiltonian recast"

$$V_{ee} = \frac{1}{2L} \sum_{k, k', q} V_{ee}(q) a_{k-q}^\dagger a_{k'+q}^\dagger a_{k'} a_k$$



"Second quantization"

$$= \frac{1}{2L} \sum_{qs} [g_4 \hat{p}_{sq} \hat{p}_{s-q} + g_2 \hat{p}_{sq} \hat{p}_{s-q}]$$

$$F.T. = \int_{-\infty}^{\infty} V_{ee} e^{-i\omega t} dt$$

$$= \frac{1}{2L} \int_{-\infty}^{\infty} [g_4 \hat{p}_{sq} \hat{p}_{s-q} + g_2 \hat{p}_{sq} \hat{p}_{s-q}] e^{-i\omega t} dt$$

$$= \frac{1}{2L} \int_{-\infty}^{\infty} [g_4 \hat{p}_{sq} \hat{p}_{s-q} + g_2 \hat{p}_{sq} \hat{p}_{s-q}] e^{i k x} dx$$

Summation $\sum_{kk'q}$

$$(k, k', q) \simeq (\pm k_F, \pm k_F, 0)$$

$$V_{ee} = \frac{1}{2L} \sum_{kk'q} V_{ee}(0) a_{k_F}^\dagger a_{k_F}^\dagger a_{k_F} a_{k_F}$$

$$(k, k', q) \simeq (\pm k_F, \mp k_F, 2k_F)$$

$$V_{ee} = \frac{1}{2L} \sum_{kk'q} V_{ee}(2k_F) a_{k_F}^\dagger a_{k_F}^\dagger a_{k_F} a_{k_F}$$

Exercise #11:

(2.36) "Commutator algebra"

$$[\hat{A}, \hat{B}\hat{C}] = [\hat{A}, \hat{B}] \hat{C} + \hat{B} [\hat{A}, \hat{C}]$$

(pg 6.0) "Auxiliary identity to Commutation relation"

$$\begin{aligned}
[\hat{p}_{sq}, \hat{p}_{s'q'}] &= \delta_{ss'} \sum_k (a_{sk+q}^\dagger a_{sk-q'} - a_{sk+q+q'}^\dagger a_{sk}) \\
&= \delta_{ss'} \sum_k (a_{sk+q}^\dagger a_{sk+q} a_{-q}^\dagger a_{-q'} - a_{sk}^\dagger a_{q'}^\dagger a_{q'} a_{sk}) \\
&= \delta_{ss'} \sum_k (\hat{n}_{sk+q} a_{-q}^\dagger a_{-q'} - \hat{n}_{sk} a_{q'}^\dagger a_{q'}) \\
&= \delta_{ss'} \sum_k \langle \Omega | \hat{n}_{sk+q} a_{q'}^\dagger a_{-q} - \hat{n}_{sk} a_{q'}^\dagger a_{-q} | \Omega \rangle
\end{aligned}$$

$$= \delta_{ss'} \delta_{q, -q'} \sum_k \langle \Omega | \hat{n}_{sk+q} - \hat{n}_{sk} | \Omega \rangle$$

... Where $\hat{n}_{sk} = a_{sk}^\dagger a_{sk}$
 $\langle \Omega | a_{sk}^\dagger a_{sk} | \Omega \rangle = \delta_{k0}$

Real observable: energy difference,
 transition, or
 ground state.

Exercise #12: Expectation values:

• Wavefunction - $|\psi_{sq}\rangle \equiv \hat{p}_{sq} |\Omega\rangle$

◦ Operator - $\hat{H}_0 = \hat{H}_0' + \text{const}$

◦ Transition - $\langle 4f_{sq} | \hat{H}_0 | 4f_{sq} \rangle = \langle 4f_{sq} | \hat{H}_0' + \text{const} | 4f_{sq} \rangle$
 $= \langle 4f_{sq} | \hat{H}_0' | 4f_{sq} \rangle + \langle 4f_{sq} | \text{const} | 4f_{sq} \rangle$
 $= \langle 4f_{sq} | \hat{H}_0' | 4f_{sq} \rangle + \text{const} \langle \Omega | \hat{P}_{sq} \hat{P}_{sq} | \Omega \rangle$
 $= \langle 4f_{sq} | \hat{H}_0' | 4f_{sq} \rangle$

Exercise #13:

Expectation values:

◦ Operator - $\hat{S}' \equiv \sum_m \hat{S}'_m$

◦ Wavefunction - $|\alpha\rangle \equiv \exp(i\alpha \cdot \hat{S}) |\Omega\rangle$

◦ Transition - $\langle \alpha | \hat{S}' | \alpha \rangle = \langle \Omega | e^{-i\alpha \cdot \hat{S}} \cdot \hat{S}' \cdot e^{i\alpha \cdot \hat{S}} | \Omega \rangle$
 $= \sum_m S'_m \langle \Omega | \Omega \rangle$
 $= \sum_m S'_m$

Energy Expectation:

$$\langle \Omega | \Omega \rangle = \int_{-\infty}^{\infty} S_m \otimes_m \hat{H} \otimes_m S_m d\tau$$

$$= \frac{1}{N} \int_{-\infty}^{\infty} S_m^2 d\tau$$

(pg 74)
 "Ground state"
 $|\Omega\rangle = \otimes_m |S_m\rangle$

$$= \frac{E}{N} \text{ "average energy"}$$

State $|\pi/2, 0, 0\rangle$:

(pg)

If $\alpha = (\pi/2, 0, 0)$, then

$$\begin{aligned}\langle \alpha | \alpha \rangle &= \langle \Omega | e^{-i\alpha s} \cdot e^{i\alpha s} | \Omega \rangle \\ &= \langle S_m | \otimes e^{-i\pi/2 s} \cdot e^{i\pi/2 s} \otimes | S \rangle\end{aligned}$$

Quantum Notation

• Wave function at a value

$$\langle 0.25 | \psi \rangle = \psi(0.25)$$

• Outer product

$$|\phi \rangle \langle \psi| = \frac{|\phi_1\rangle + |\phi_2\rangle}{\sqrt{2}} \frac{|\psi_1\rangle + |\psi_2\rangle}{\sqrt{2}}$$

• Schrodinger equation

$$\hat{H}\psi = E\psi$$

• Expectation

$$\langle \psi | \hat{H} | \psi \rangle = \langle H \rangle$$

• Superposition

$$\langle \phi | = \frac{\langle \phi_1 | + \langle \phi_2 |}{2}$$

• Inner product

$$\langle \psi | \phi \rangle = \text{Dot product.}$$

Exercise #14

(2.46) "Spin raising and lowering operators"

$$[\hat{S}_m^z, \hat{S}_n^\pm] = \pm \delta_{mn} \hat{S}_m^\pm \quad [\hat{S}_m^+, \hat{S}_n^-] = 2\delta_{mn} \hat{S}_m^z$$

(pg 75) "Holstein-Primakoff Transformation"

$$\hat{S}_m^- = a_m^\dagger (2S - a_m^\dagger a_m)^{1/2}$$

$$\hat{S}_m^+ = (2S - a_m^\dagger a_m)^{1/2} a_m$$

$$\hat{S}_m^z = S - a_m^\dagger a_m$$

Identities from Quantum Mechanics:

① Orbital Angular Momentum:

$$[S_x, S_y] = i\hbar S_z, \quad [S_y, S_z] = -i\hbar S_x, \quad [S_z, S_x] = i\hbar S_y$$

② Spin Angular Momentum:

$$S^2 = S_x^2 + S_y^2 + S_z^2$$

$$S^2 - S_z^2 = S_x^2 + S_y^2$$

$$= (S_x + iS_y)(S_x - iS_y)$$

$$= S_+ \circ S_-$$

$$[S^2, \hat{S}_\pm] = [S^2, S_x \pm iS_y]$$

$$= [S^2, S_x] \pm [S^2, iS_y]$$

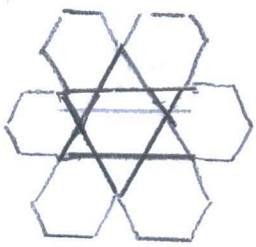
$$= 0$$

$$\begin{aligned} [S_z, \hat{S}_{\pm}] &= S_z \circ S_{\pm} - S_{\pm} \circ S_z \\ &= [S_z, S_x] \pm i[S_z, S_y] \\ &= [S_z, S_x] \pm [S_z, S_y] \\ &= i\hbar S_y \pm i(-i\hbar S_x) \\ &= \hbar(S_x \pm iS_y) \\ &= \hbar S_{\pm} \end{aligned}$$

$$\begin{aligned} [S_m^z, S_n^{\pm}] &= [S_m^z, S_n^1 \pm iS_n^2] \\ &= [S_m^z, S_n^1] \pm i[S_m^z, S_n^2] \\ &= i\hbar S_m^2 \pm i(-i\hbar S_m^1) \\ &= \hbar(S_m^1 \pm S_m^2) \\ &= \pm \hbar S_m^{\pm} \\ &= \pm \delta_{mn} \delta_m^{\pm} \end{aligned}$$

← "Book removed $\hbar$ as
with Chapter 1 or on
page 74"

$$\begin{aligned} [S_m^+, S_n^-] &= [S_m^1 + iS_m^2, S_n^1 - iS_n^2] \\ &= S_m^- a_m^+ a_n + S_m^- a_m^+ a_n \\ &= 2\delta_{mn} S_m^- \end{aligned}$$



"kagomé
lattice"

Kagomé anti ferromagnet:

Continuous spin degeneracy:

$$\phi_1 = \phi_2 + \phi$$

$$\phi_2 = \phi_3 + \phi$$

$$\phi_3 = \phi + \phi \quad \text{where } \phi = 2\pi/3$$

$$\hat{H} = J \sum \hat{S}_m \cdot \hat{S}_n$$

$$= J S^2 (\cos \theta_1 \cos \phi_1 + \cos \theta_2 \cos \phi_2 + \cos \theta_3 \cos \phi_3)$$

$$= -\frac{3}{2} J S^2$$

Problem 2.4.1: Stone-Von Neumann Theorem:

a) Operator - $n_\lambda = a_\lambda^\dagger a_\lambda$

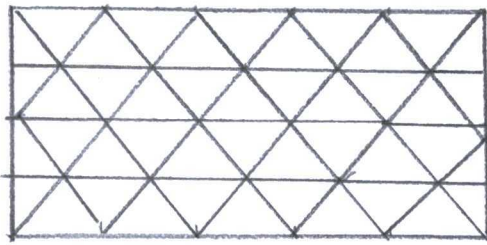
Wave function $|n_{\lambda_1}, n_{\lambda_2}, \dots\rangle$

Eigenvalues $\hat{n} |n_{\lambda_1}, n_{\lambda_2}, \dots\rangle = n |n_{\lambda_1}, n_{\lambda_2}, \dots\rangle$

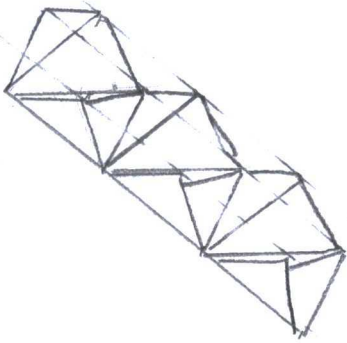
Unique basis when a derivative provides
no change to eigenvalues:

$$|n_{\lambda_1}, n_{\lambda_2}, \dots\rangle = P([a_\mu, a_\mu^\dagger]) |n_{\lambda_1}, n_{\lambda_2}, \dots\rangle$$

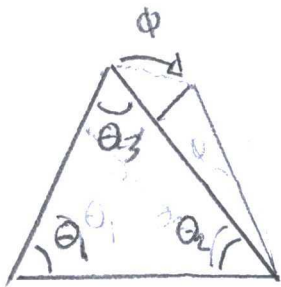
Exercise #15.



"Triangular
lattice"



"Equilateral
triangular
pyramids
lattice"



"equilateral
triangle
rotated"

(pg 76) "Heisenberg
Antiferromagnet"

$$\hat{H} = J \sum_{\langle mn \rangle} \hat{S}_m \cdot \hat{S}_n$$

$$= JS^2 \sum_{\langle mn \rangle} S_i S_m$$

$$= JS^2 (\cos \theta_1 \cos \phi_1 + \cos \theta_2 \cos \phi_2 + \cos \theta_3 \cos \phi_3)$$

$$\therefore \text{When } \theta_1 = \frac{2\pi}{3} \quad \theta_2 = \frac{2\pi}{3} \quad \theta_3 = \frac{2\pi}{3}$$

$$\phi_1 = 0 \quad \phi_2 = \frac{2\pi}{3} \quad \phi_3 = \frac{\pi}{2}$$

$$= -\frac{3}{2} JS^2$$

Final Energy:

Schrodinger equation:

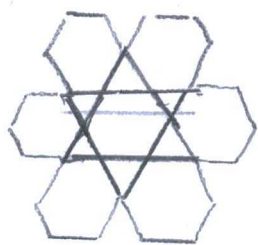
$$\hat{H} \psi = -\frac{3}{2} JS^2 \psi$$

$$= E \psi$$

$$E = -\frac{3}{2} JS^2$$

Fun fact:

Book describes
Ferromagnetic
as negative
energy and
antiferromagnetic
as positive.



"Kagomé
lattice"

Kagomé anti ferromagnets

Continuous spin degeneracy:

$$\phi_1 = \phi_2 + \phi$$

$$\phi_2 = \phi_3 + \phi$$

$$\phi_3 = \phi + \phi \quad \text{... where } \phi = 2\pi/3$$

$$\hat{H} = J \sum \hat{S}_m \cdot \hat{S}_n$$

$$= J S^2 (\cos \theta_1 \cos \phi_1 + \cos \theta_2 \cos \phi_2 + \cos \theta_3 \cos \phi_3)$$

$$= -\frac{3}{2} J S^2$$

Problem 2.4.1: Stone-Von Neumann Theorem:

a) Operator - $n_\lambda = a_\lambda^\dagger a_\lambda$

Wave function $|n_{\lambda_1}, n_{\lambda_2}, \dots\rangle$

Eigenvalues $\hat{n} |n_{\lambda_1}, n_{\lambda_2}, \dots\rangle = n |n_{\lambda_1}, n_{\lambda_2}, \dots\rangle$

Unique basis when a derivative provides
no change to eigenvalues:

$$|n_{\lambda_1}, n_{\lambda_2}, \dots\rangle = P([a_\mu, a_\mu^\dagger]) |n_{\lambda_1}, n_{\lambda_2}, \dots\rangle$$

$$= P[\hat{n}] |n_{\lambda_1}, n_{\lambda_2}, \dots\rangle$$

$$= n_{\lambda} |n_{\lambda_1}, n_{\lambda_2}, \dots\rangle$$

$$b) \hat{n}_{\lambda} a_{\lambda} |n_{\lambda_1}, n_{\lambda_2}, \dots\rangle = (n-1) a_{\lambda} |n_{\lambda_1}, n_{\lambda_2}, \dots\rangle$$

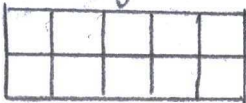
Where $n-1 \geq 0$ and $n \geq 1$

Problem 2.4.2: Semi-classical spin waves:

Lattices:



"Triangular"



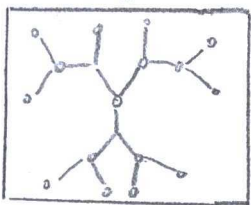
"Square"



"Kagomé"



"Honeycomb"



"Bethe"

a) (from Problem) "spin commutator relation"

$$[\hat{S}_i^{\alpha}, \hat{S}_j^{\beta}] = i \delta_{ij} \epsilon^{\alpha\beta\gamma} \hat{S}_i^{\gamma}$$

(from Problem) "Operator identity"

$$i \hat{S}_i = [\hat{S}_i, \hat{H}]$$

(2.44) "Heisenberg ferromagnet"

$$\hat{H} = -J \sum_{\langle mn \rangle} \hat{S}_m \cdot \hat{S}_n$$

$$= -J \sum_{\langle mn \rangle} \hat{S}_m \cdot \hat{S}_{m+1} \quad \dots \text{for one-dimensional ferromagnet}$$

$$\begin{aligned} [\hat{S}_i, \hat{H}] &= [\hat{S}_i, -J \sum \hat{S}_m \cdot \hat{S}_{m+1}] \\ &= i \hat{S}_i \end{aligned}$$

Commutator relations:

$$[\hat{S}_i^{\alpha}, \hat{S}_i^{\beta} \hat{S}_{i+1}^{\beta}] = [\hat{S}_i^{\alpha}, \hat{S}_i^{\beta}] \hat{S}_{i+1}^{\beta} + \hat{S}_i [\hat{S}_i^{\alpha}, \hat{S}_{i+1}^{\beta}]$$

$$= i \epsilon^{\alpha\beta\gamma} \cdot s_i^\alpha \cdot s_{i+1}^\beta$$

$$[s_i^\alpha, s_{i-1}^\beta \cdot s_i^\gamma] = [s_i^\alpha, \hat{s}_{i-1}^\beta] \hat{s}_i^\gamma + \hat{s}_{i-1}^\beta [s_i^\alpha, \hat{s}_i^\gamma]$$

$$= i \epsilon^{\alpha\beta\gamma} \cdot (\hat{s}_{i-1} \cdot \hat{s}_i)_\alpha$$

$$[\hat{s}_i, \hat{H}_{\text{TOT}}] = [\hat{s}_i, \hat{H}_{i-1} + \hat{H}_{i+1}]$$

$$= [\hat{s}_i, -J \sum \hat{s}_{i-1} \cdot \hat{s}_i - J \sum \hat{s}_i \cdot \hat{s}_{i+1}]$$

$$= [\hat{s}_i, -J \sum s_{i-1} \cdot s_i] + [\hat{s}_i, -J \sum s_i \cdot s_{i+1}]$$

$$= -iJ [(s_{i-1} \cdot s_i) + (s_i \cdot s_{i+1})]$$

$$= -i \hat{s}_i$$

$$\hat{s}_i = J \cdot s_i \times (s_{i+1} + s_{i-1})$$

b) Equation of Motion:

$$\hat{s}_i = J \cdot s_i \times (s_{i+1} + s_{i-1})$$

$$= J \cdot s_i \times (s_i + \partial_x s_i + \frac{1}{2} \partial^2 s_i + s_i - \partial_x s_i + \frac{1}{2} \partial^2 s_i)$$

$$= J \cdot s_i \times (2s_i + \partial^2 s_i)$$

$$[s_i, s_i] = 0$$

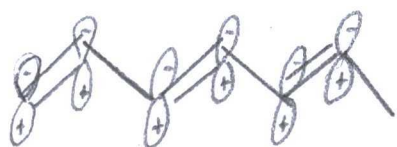
$$= J \cdot s_i \times \partial^2 s_i$$

oo edition no. 2 versus no. 3 mayhem!

Problem 2.4.3: Su-Schrieffer-Heeger model of a conducting polymer chain:



"Polyacetylene"



"sp²-hybridized polymer chain"



"Peierls distorted chain"

a) (from problem) "Effective Hamiltonian"

$$\hat{H} = -t \sum_{n=1, \sigma}^N (1 + u_n) [c_{n\sigma}^\dagger c_{n+1\sigma} + h.c.] + \sum_{n=1}^N \frac{Ks}{2} (u_{n+1} - u_n)^2$$

(from problem) "Lattice distortion"

$$u_n = (-1)^n \alpha$$

$$\hat{H}_{TOT} = \hat{H}_{n=even} + \hat{H}_{odd}$$

$$= -t \sum_{m=1, \sigma}^{N/2} \{ (1 + \alpha) [a_{m\sigma}^\dagger b_{m\sigma} + h.c.] + (1 - \alpha) [b_{m\sigma}^\dagger a_{m+1\sigma} + h.c.] \} + 2NRs \alpha^2$$

(2.24) "Operator Transformation"
[Block and Wannier states]

$$a_{k\sigma}^\dagger = \frac{1}{\sqrt{N}} \sum_i e^{ikR_i} a_{i\sigma}^\dagger$$

$$a_{i\sigma}^\dagger = \frac{1}{\sqrt{N}} \sum_k e^{-ikR_i} a_{k\sigma}^\dagger$$

... Where $a_i = \sqrt{2}$

$$\hat{H} = -t \sum_{m=1, \sigma}^{N/2} \left\{ (1+\alpha) \left[\sum_k \frac{z}{N} e^{2ikm} + h.c. \right] + (1-\alpha) \left[\sum_k \frac{z}{N} e^{2ikm} + h.c. \right] \right\} + 2Nk\alpha^2$$

$$= -t \sum_k \begin{pmatrix} a_{k\sigma}^\dagger & b_{k\sigma}^\dagger \end{pmatrix} \begin{pmatrix} 0 & (1+\alpha)+(1-\alpha)e^{2ikm} \\ (1+\alpha)+(1-\alpha)e^{-2ikm} & 0 \end{pmatrix} \begin{pmatrix} a_{k\sigma} \\ b_{k\sigma} \end{pmatrix}$$

Eigen spectrum:

① Matrix:

$$\hat{H} = \begin{pmatrix} 0 & (1+\alpha)+(1-\alpha)e^{2ikm} \\ (1+\alpha)+(1-\alpha)e^{-2ikm} & 0 \end{pmatrix}$$

② Eigenvalues:

$$\hat{H} - E = \begin{pmatrix} 2Nk\alpha^2 - E & 2Nk\alpha^2 + (1+\alpha)+(1-\alpha)e^{2ikm} \\ 2Nk\alpha^2 + (1+\alpha)+(1-\alpha)e^{-2ikm} & 2Nk\alpha^2 - E \end{pmatrix}$$

= 0

$$E_1 = +2\sqrt{1+(\alpha^2-1)\sin^2 k} - 2Nk\alpha^2$$

$$E_2 = -2\sqrt{1+(\alpha^2-1)\sin^2 k} - 2Nk\alpha^2$$

③ Eigenvectors:

... "on a time limit"

④ Eigenvectors are linearly independent.

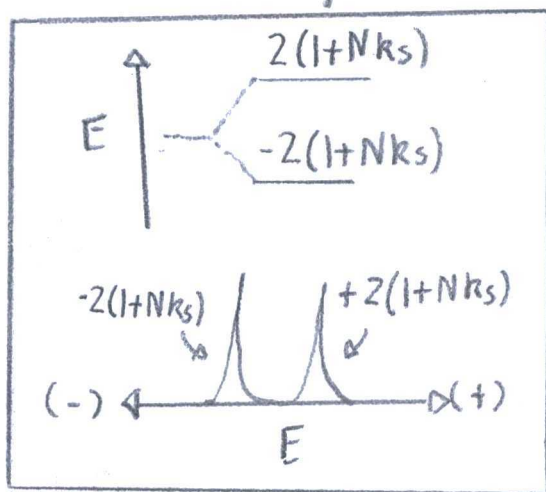
$$E_1 = 2\sqrt{1 + (\alpha^2 - 1)\sin^2 k} - 2Nk_s \alpha^2$$

$$E_2 = -2\sqrt{1 + (\alpha^2 - 1)\sin^2 k} - 2Nk_s \alpha^2$$

Distortion near Zero:

$$\lim_{\alpha \rightarrow 0} E = \lim_{\alpha \rightarrow 0} \left[\pm 2\sqrt{1 + (\alpha^2 - 1)\sin^2 k} \right] - 2Nk_s \alpha^2$$

$= \pm 2(1 + Nk_s)$ "A gap in the spectrum"



b)

(from Problem) "Formula"

$$\int_{-\pi/2}^{\pi/2} dk (1 - (1 - \alpha^2)\sin^2 k)^{1/2} \simeq 2 + (a_1 - b_1 \ln \alpha^2) \alpha^2 + O(\alpha^2 \ln^2 \alpha^2)$$

$$\int_{-\pi/2}^{\pi/2} E dk = \int_{-\pi/2}^{\pi/2} \pm 2t \sqrt{1 + (\alpha^2 - 1) \sin^2 k} - 2NR_s \alpha^2 dk$$

$$= \pm 2t [2 + (a_1 - b_1 \ln \alpha^2) \alpha^2] - 2NR_s \alpha^2$$

Maximum distortion:

$$\frac{dE}{d\alpha} = \frac{d}{d\alpha} [\pm 2t [2 + (a_1 - b_1 \ln \alpha^2)] - 2NR_s \alpha^2]$$

$$= \pm 2t [2a_1 \alpha - b_1 \left[\frac{2}{\alpha} \alpha^2 + 2\alpha^2 \ln \alpha^2 \right] - 4NR_s \alpha]$$

$$= 0$$

$$\alpha = e^{\frac{1}{2}(a_1/b_1 - 1 - NR_s/tb_1)}$$

Notes Peierls describes
two spectra by two
structures in a molecule.

c) The Peierls instability has two degenerate configurations: conjugated and unconjugated. An odd site number alters the structure and spectrum.

Problem 2.4.4: Schwinger boson representation:

(from problem) "Schwinger Boson representation"

$$\hat{S}^+ = a^\dagger b \quad \hat{S}^- = (\hat{S}^+)^\dagger \quad \hat{S}^z = \frac{1}{2}(a^\dagger a - b^\dagger b)$$

Commutation relation:

$$\begin{aligned} [\hat{S}^+, \hat{S}^-] &= \hat{S}^+ \hat{S}^- - \hat{S}^- \hat{S}^+ \\ &= a^\dagger b \cdot (\hat{S}^+)^\dagger - (\hat{S}^+)^\dagger a^\dagger b \\ &= a^\dagger b - b^\dagger b \\ &= 2 \cdot \hat{S}^z \end{aligned}$$

$$\begin{aligned} \text{b) If } \hat{S}^2 &= (\hat{S}^z)^2 + \frac{1}{2}(\hat{S}^+ \hat{S}^- + \hat{S}^- \hat{S}^+) \\ &= \frac{1}{4}(a^\dagger a - b^\dagger b)^2 + \frac{1}{2}(\hat{n}_a \cdot \hat{n}_b + \hat{n}_b n_a) \\ &= \frac{1}{2}(\hat{n}_a + \hat{n}_b)^2 \end{aligned}$$

Total Spin:

$$\hat{S}^2 |s, m\rangle = s(s+1) |s, m\rangle$$

(from problem)

$$n_a + n_b |n_a, n_b\rangle = 2s$$

$$\hat{S}^2 |s, m\rangle = \frac{1}{4}(\hat{n}_a + \hat{n}_b)^2 |s, m\rangle$$

Spin operator:

$$\hat{S}^z |s, m\rangle = \frac{1}{2}(\hat{n}_a - \hat{n}_b) |n_a = s, n_b = s - m\rangle \\ = m |s, m\rangle$$

Problem 2.4.5: Jordan-Wigner Transformation:

(from problem) "Spin lowering and raising operators"

$$\hat{S}^+ = f^\dagger ; \hat{S}^- = f ; \hat{S}^z = f^\dagger f - 1/2$$

a) Commutator relationship:

$$[\hat{S}^+, \hat{S}^-] = \hat{S}^+ \hat{S}^- - \hat{S}^- \hat{S}^+ \\ = f^\dagger f - f f^\dagger \\ = 2S^z$$

Nonlinear Operator Transformation:

$$\hat{S}_e^+ = f_e^\dagger \exp(i\pi \sum_{j < e} \hat{n}_j)$$

$$\hat{S}_e^- = \exp(-i\pi \sum_{j < e} \hat{n}_j) f_e$$

$$\hat{S}_e^z = f_e^\dagger f_e - 1/2$$

$$[\hat{S}_e^+, \hat{S}_e^-] = \hat{S}_e^+ \hat{S}_e^- - \hat{S}_e^- \hat{S}_e^+$$

$$= f_e^\dagger \exp(+i\pi \sum_{j < e} \hat{n}_j) \exp(-i\pi \sum_{j < e} \hat{n}_j) f_e - \exp(-i\pi \sum_{j < e} \hat{n}_j)$$

$$\cdot f_e \circ f_e^\dagger \exp(+i\pi \sum_{j < e} \hat{n}_j)$$

Real-world

application:

Spin operators
apply to fermions
in Electron magnetic

Paramagnetic
Resonance and
bosons in Nuclear
magnetic resonance.

$$\begin{aligned}
&= f_e^\dagger f_e - f_e^\dagger f^\dagger \\
&= 2f^\dagger f - 1 \\
&= 2S^z
\end{aligned}$$

$$\begin{aligned}
b) \hat{S}_m^+ \hat{S}_{m+1}^- &= f_m^\dagger \exp(i\pi \sum_{j=m} n_j) \exp(-i\pi \sum_{j=m+1} n_j) f_{m+1} \\
&= f_m^\dagger e^{-i\pi n_m} \circ f_{m+1} \\
&= f_m^\dagger \circ f_{m+1}
\end{aligned}$$

... when $n=2$ particles

c) (from problem) "Heisenberg Spin Chain"

$$\begin{aligned}
\hat{H} &= \sum_n \left(J_z \hat{S}_n^z \hat{S}_{n+1}^z + \frac{J_\perp}{2} (\hat{S}_n^+ \hat{S}_{n+1}^- + \hat{S}_n^- \hat{S}_{n+1}^+) \right) \\
&= - \sum_n \left(J_n \left(f_n^\dagger f_n - \frac{1}{2} \right) \left(f_{n+1}^\dagger f_{n+1} - \frac{1}{2} \right) + \frac{J_\perp}{2} (f_n^\dagger \circ f_{n+1}) \right. \\
&\quad \left. + f_{n+1}^\dagger \circ f_n \right) \\
&= - \sum_n \left(\frac{J_\perp}{2} (f_n^\dagger \circ f_{n+1} + h.c.) + J_z \left(\frac{1}{4} - f_n^\dagger \circ f_n \right. \right. \\
&\quad \left. \left. + f_n^\dagger f_n f_{n+1}^\dagger f_{n+1} \right) \right)
\end{aligned}$$

d) If $J_z = 0$, then

$$\begin{aligned}
\hat{H} &= - \sum_n \left(\frac{J_\perp}{2} (f_n^\dagger f_{n+1} + h.c.) \right) \\
&= - \sum_n \frac{J_\perp}{2} (f_n^\dagger f_{n+1}) \begin{pmatrix} 0 & e^{-2\pi i a} \\ e^{-2\pi i a} & 0 \end{pmatrix} \begin{pmatrix} f_n \\ f_{n+1} \end{pmatrix}
\end{aligned}$$

Eigen spectrum:

① Matrix: $\hat{H} = \begin{pmatrix} 0 & -2\pi i a \\ -2\pi i a & 0 \end{pmatrix}$

② Eigenvalues:

$$(\hat{H} - E) = \begin{pmatrix} -E & -2\pi i a \\ -2\pi i a & -E \end{pmatrix}$$

$$= 0$$

$$E_1 = e^{-2\pi i a}$$

$$E_2 = -e^{-2\pi i a}$$

③ Eigenvectors:

$$(\hat{H} - E_1) = \begin{pmatrix} -2\pi i a & -2\pi i a \\ -e & e \\ -2\pi i a & -e \\ e & -e^{-2\pi i a} \end{pmatrix}$$

$$(\hat{H} - E_1)\psi = \begin{pmatrix} -2\pi i a & -2\pi i a \\ -e & e \\ -2\pi i a & -e \\ e & -e \end{pmatrix} \begin{pmatrix} |\psi\rangle \\ |\psi\rangle \end{pmatrix}$$

$$= \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$|\psi\rangle = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

$$(\hat{H} - E_2)\psi = \begin{pmatrix} -2\pi i a & -2\pi i a \\ e & e \\ -2\pi i a & -e \\ e & e \end{pmatrix} |\psi\rangle$$

$$= 0$$

$$|\psi\rangle = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\hat{H}|\psi\rangle = E|\psi\rangle$$

$$E = -J \cos(Ra) \quad \uparrow$$

④ Eigenvectors are linearly independent

$$R_0 / R_1 = \cos(2\pi b_1) = \cos(Ra)$$

Problem 2.4.6: Spin-charge separation in one-dimension

Hydrodynamic
photographs
were unreal
for spin-
charge ^{super}
separation.

(2.34) "Operator"

$$\hat{p}_{sq} = \sum_k^+ a_{s(k+q)} a_{sk}$$

(from problem) "Operator"

$$\hat{p}_{sq\alpha} = \sum_k^+ a_{s(k+q)\alpha} \cdot a_{sk\alpha} \quad \dots \text{where } \alpha = \uparrow, \downarrow$$

(2.39) "bosonic commutation relation"

$$b_q = n_q \hat{p}_{L-q} \quad b_q^\dagger = n_q \hat{p}_{L-q}$$

$$b_{-q} = n_q \hat{p}_{R-q} \quad b_{-q}^\dagger = n_q \hat{p}_{Rq}$$

(from problem) "spin index"

$$b_q \rightarrow b_{q\alpha}$$

(from problem) "Momentum spin interaction"

$$|k_F + q + q_1, \uparrow; k_F + q - q_1, \downarrow\rangle$$

$$\rightarrow |k_F + q + q_z, \uparrow; k_F + q - q_z, \downarrow\rangle$$

(from problem) "Effective Bosonic Hamiltonian"

$$\hat{H} = \sum_{q>0, s, q} V_F \cdot q \cdot b_{sq\alpha}^\dagger \cdot b_{sq} + \sum_{q>0, s, \alpha\alpha'} |q| \left[\frac{g_2}{2\pi} (b_{sq\alpha}^\dagger b_{sq\alpha'}^\dagger + \text{h.c.}) + \frac{g_4}{2\pi} b_{sq\alpha}^\dagger b_{sq\alpha'} \right]$$

(from problem) "Operators' Charge"

$$b_{sqe} = \frac{1}{\sqrt{2}} (b_{sq\uparrow} + b_{sq\downarrow})$$

"operator spin"

$$b_{sq\sigma} = \frac{1}{\sqrt{2}} (b_{sq\uparrow} - b_{sq\downarrow})$$

$$\hat{H} = \sum_{q>0, s} V_F \cdot q (b_{sqe}^\dagger b_{sqe} + b_{sq\sigma}^\dagger b_{sq\sigma}) + |q| \left[\frac{g_2}{\pi} (b_{sqe}^\dagger b_{sqe}^\dagger + \text{h.c.}) + \frac{g_4}{\pi} b_{sqe}^\dagger b_{sqe} \right]$$

(pg 71) "Bogoliubov Transformation"

$$T = \begin{pmatrix} \cosh \Theta & \sinh \Theta \\ \sinh \Theta & \cosh \Theta \end{pmatrix}$$

$$\text{where } \tanh(2\Theta) = -g_2 / (2\pi V_F + g_4)$$

} "The book confused me on page 71. Wikipedia provided a formula after $\hat{H}$ -rearrangement"

$$\hat{H} = \sum_{q>0,s} \left[\underbrace{(V_F q + \frac{g_4}{\pi})}_{"u"} b_{sqe} + \underbrace{|q| \frac{g_2}{\pi}}_{"v"} b_{sqe}^\dagger \right] b_{sqe}^\dagger + V_F q b_{sq\sigma}^\dagger b_{sq\sigma}$$

Wikipedia Bogoliubov transformation:

① Transformation matrix

$$U = \begin{pmatrix} u & v \\ v^\dagger & u^\dagger \end{pmatrix}$$

② Inverse matrix:

$$U^{-1} = \begin{pmatrix} u & v^\dagger \\ v & u^\dagger \end{pmatrix}$$

③ Gamma:

$$T = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

④ Diagonalization:

$$U(T H) U^{-1} = T D$$

$$\hat{H} = \sum_{q>0,s} \left[|q| \begin{pmatrix} b_{sq} & b_{sqe}^\dagger \end{pmatrix} \begin{pmatrix} (V_F + \frac{g_4}{\pi})^2 - (\frac{g_2}{\pi})^2 & 0 \\ 0 & 0 \end{pmatrix} b_{sqe}^\dagger \right] + |q| V_F b_{sq\sigma}^\dagger b_{sq\sigma}$$

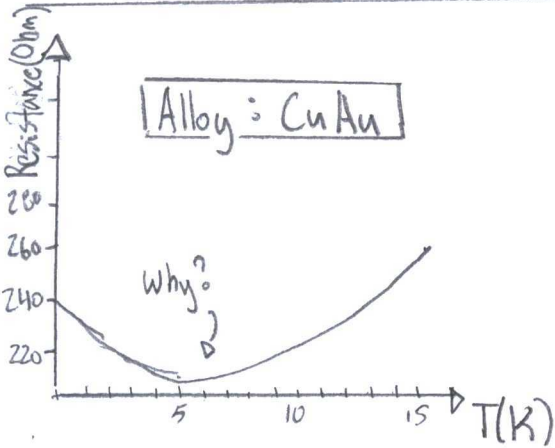
$$= \sum_{q>0,5} |q| \left(\left(V_F + \frac{g_4}{\pi} \right)^2 + \left(\frac{g_2}{\pi} \right)^2 \right) b_{sqe}^\dagger b_{sqe} + |q| V_F b_{sqo}^\dagger b_{sqo}$$

(Equation 2.7)

$$b_{sqe}^\dagger b_{sqe}^\dagger = 0$$

Pr-11

Problem 2.4.7: The Kondo Problem:



Cite: W. J. de Haas

"The electrical resistance of gold, copper, and lead at low temperatures"

Physical 1. 1115 (1934)

Function

$$\rho(T) = \rho_0 + aT^2 + c_m \ln \frac{u}{T} + bT^5$$

Energy Function

$$E_{TOTAL} = E_{mag} + E_{repulsion} + E_{impurity \text{ ratio}}$$

$$\hat{H} = H_{of} + H_{od} + (H_{corr} + H_{sd})$$

Cite: J. Kondo

"Resistance in dilute Magnetic alloys"

Prog. Theor. Phys. 32, 37 (1964)

Cite: P. W. Anderson

"Localized magnetic States in metals"

Phys Rev. 124, 41 (1961)

(from problem) "Anderson impurity Hamiltonian"

$$\hat{H} = \sum_{\mathbf{k}\sigma} (\epsilon_{\mathbf{k}} c_{\mathbf{k}\sigma}^{\dagger} c_{\mathbf{k}\sigma} + (V_{\mathbf{k}} d_{\sigma}^{\dagger} c_{\mathbf{k}\sigma} + h.c.)) + \sum_{\sigma} \epsilon_d n_{d\sigma} + U n_{d\uparrow} n_{d\downarrow}$$

Where $c_{\mathbf{k}\sigma}^{\dagger}$ = electron operator

σ = spin

$\epsilon_{\mathbf{k}}$ = metallic host

$d_{\sigma}^{\dagger}$ = operator on impurity.

Again, the problem relates minimum energy to spin, orbitals, and Coulomb repulsion from boundary conditions in an amalgam.

$$V_{\mathbf{k}} = e^{i\mathbf{k}\cdot\mathbf{d}} \langle \phi_d | \hat{H} | \psi_d \rangle = \text{coupling between states.}$$

ϕ_d = atomic d-level

ψ_d = Conducting d-level

$n_{d\sigma} = d_{\sigma}^{\dagger} d_{\sigma}$ = number operator

U = Coulomb interaction strength

$\epsilon_d < \epsilon_F < \epsilon_d + U$ - Fermi level

Minimum Hamiltonian:

Schrodinger equation: $\sum_{n=0}^Z \hat{H}_{mn} |\psi_n\rangle = E |\psi_n\rangle$

Hamiltonian: $\hat{H}_{mn} = \hat{P}_m \hat{H} \hat{P}_n$

Projection

Where P_m = Projection onto m-electron

$$\hat{P}_0 = \prod_{\sigma} (1 - n_{d\sigma})$$

$$\hat{P}_1 = (1 + n_{d\uparrow} + n_{d\downarrow} - 2n_{d\uparrow}n_{d\downarrow})$$

$$\hat{P}_2 = n_{d\uparrow} n_{d\downarrow}$$

a) $\hat{H}_{mn} = \hat{P}_m \hat{H} \hat{P}_n$

$$= \hat{P}_m \begin{pmatrix} E_K C_{K\sigma}^\dagger C_{K\sigma} & V_d d_\sigma^\dagger (1 - n_d) C_{K\sigma} & 0 \\ V_d d_\sigma (1 - n_{d\sigma}^\dagger) C_{K\sigma}^\dagger & E_K C_{K\sigma}^\dagger C_{K\sigma} + E_d & V_K d_\sigma^\dagger n_{d\sigma} C_{K\sigma} \\ 0 & V_K d_\sigma n_{d\sigma}^\dagger C_{K\sigma}^\dagger & E_K C_{K\sigma}^\dagger C_{K\sigma} + 2E_d + U \end{pmatrix} \hat{P}_n$$

Coupling involves $n=1, m=1$

electrons in $\hat{H}_{mn} = \hat{P}_m \begin{pmatrix} H_{00} & H_{01} & H_{02} \\ H_{10}^\dagger & H_{11} & H_{12} \\ H_{20}^\dagger & H_{21}^\dagger & H_{22} \end{pmatrix} \hat{P}_n$

So, $\hat{H}_{20} = \hat{H}_{02} = 0$

While $\hat{H}_{12} = V_K d_\sigma^\dagger n_{d\sigma} C_{K\sigma}$

$\hat{H}_{10} = V_d d_\sigma^\dagger (1 - n_d) C_{K\sigma}$

b) Schrodinger Equation

$$H_{mn} |2f\rangle = E |2f\rangle$$

$$\begin{pmatrix} \hat{H}_{00} & \hat{H}_{01} & 0 \\ \hat{H}_{10} & \hat{H}_{11} & \hat{H}_{12} \\ 0 & \hat{H}_{21} & \hat{H}_{22} \end{pmatrix} \begin{pmatrix} |2f_0\rangle \\ |2f_1\rangle \\ |2f_2\rangle \end{pmatrix} = E \begin{pmatrix} |2f_0\rangle \\ |2f_1\rangle \\ |2f_2\rangle \end{pmatrix}$$

$$\hat{H}_{00} |2f_0\rangle + \hat{H}_{01} |2f_1\rangle = E |2f_0\rangle$$

$$|2f_0\rangle = \frac{\hat{H}_{01} |2f_1\rangle}{E - H_{00}}$$

$$\hat{H}_{21} |2f_1\rangle + \hat{H}_{22} |2f_2\rangle = E |2f_2\rangle$$

$$|2f_2\rangle = \frac{\hat{H}_{21} |2f_1\rangle}{E - H_{22}}$$

$$H_{10} |2f_0\rangle + H_{11} |2f_1\rangle + H_{12} |2f_2\rangle = E |2f_1\rangle$$

$$\left[\hat{H}_{10} \frac{1}{E - H_{00}} H_{01} + H_{11} + H_{12} \frac{1}{E - H_{22}} H_{21} \right] |2f_1\rangle = E |2f_1\rangle$$

$$b) \hat{H}_{12} = \hat{H}_{12} \frac{1}{E - H_{22}} \hat{H}_{21}$$

$$= \sum_{\mathbf{k}, \sigma, \sigma'} V_{\mathbf{k}} V_{\mathbf{k}'}^* c_{\mathbf{k}\sigma} n_{\mathbf{k}\sigma'} \frac{1}{E - \epsilon_{\mathbf{k}} + 2\epsilon_{\mathbf{k}'} - U} d_{\sigma}^{\dagger} n_{d\sigma} c_{\mathbf{k}\sigma'}$$

$$\hat{H}_{10} = \hat{H}_{10} \frac{1}{E - H_{00}} \hat{H}_{01}$$

$$= \sum_{k h' \sigma'} V_k V_k d_{\sigma'}^{\dagger} (1 - n_{d\sigma'}) c_{k\sigma'} \frac{1}{E - \epsilon_k} c_{k\sigma'}^{\dagger} (1 - n_{d\sigma'}) d_{\sigma'}$$

$$\hat{H} = \sum_{k h' \sigma'} V_k V_k \left(\frac{c_{k'\sigma'} c_{k'\sigma'}^{\dagger} d_{\sigma'} d_{\sigma'}^{\dagger}}{U + \epsilon_d - \epsilon_{k'}} + \frac{c_{k'\sigma'}^{\dagger} c_{k\sigma'} d_{\sigma'}^{\dagger} d_{\sigma'}}{\epsilon_k - \epsilon_d} \right)$$

∞ when $\epsilon_k = E$

$$\simeq \sum_{k h' \sigma'} V_k V_{k'} \left(2 \hat{S}_{kh} \cdot \hat{S}_d + \frac{1}{2} \sum_{\sigma} c_{k\sigma}^{\dagger} c_{k'\sigma} n_{d\sigma} \right) \times \left(\frac{1}{U + \epsilon_d - \epsilon_{k'}} - \frac{1}{\epsilon_k - \epsilon_d} \right)$$

$$\simeq \sum_{k h'} \left[2 \cdot J_{kh} \cdot \hat{S}_{kh} \cdot \hat{S}_d + K_{kh'} \sum_{\sigma} c_{k\sigma}^{\dagger} c_{k'\sigma} \right]$$

Where $J_{kh'} = \text{Exchange interaction}$

$K_{kh'} = \text{Scattering term}$

$$\hat{H}_{sd} = \sum_{k\sigma} \epsilon_k c_{k\sigma}^{\dagger} c_{k\sigma} + \hat{H}$$

$$= \sum_{k\sigma} \epsilon_k c_{k\sigma}^{\dagger} c_{k\sigma} + 2 \cdot J \hat{S}_{kh} \cdot \hat{S}_d$$

∞ When Scattering ($K_{kh'} = 0$)